

Othmane Echchabi

Ph.D. Student, Computer Science, McGill University & Mila
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RESEARCH INTERESTS

Geospatial foundation models, self-supervised and temporal representation learning, and multimodal fusion for Earth observation. I am interested in a specific gap: foundation models transfer well to semantic tasks such as crop-type and land-cover classification, but degrade on the time-dependent targets that decision-makers actually need—crop yield, wildfire, and flood prediction—where supervised baselines often remain competitive, and in what pretraining would have to look like to close it. Applications throughout to climate risk, food security, and environmental monitoring.

EDUCATION

McGill University & Mila – Quebec AI Institute

Ph.D. in Computer Science, advised by Prof. David Rolnick

Montréal, QC

2026 – Expected 2030

- Fast-tracked from the M.Sc. program (GPA 4.0/4.0), Aug. 2025 – 2026.

Duke University / Duke Kunshan University

B.S. in Data Science

Durham, NC / Suzhou, China

Aug. 2021 – May 2025

- Signature Work (thesis): [Monitoring *Spartina alterniflora* Using Self-Supervised Learning](#).
- Study abroad: Universidad Carlos III de Madrid, Spain (Spring 2022).

PUBLICATIONS

1. Y. Zhang*, **O. Echchabi***, T. Feng, W. Zhang, H.-K. Liao, C. Chang. *Detecting Transportation Mode Using Dense Smartphone GPS Trajectories and Transformer Models*. **International Journal of Geographical Information Science** (2026). [Paper](#) | [arXiv](#) | [Code](#).
2. **O. Echchabi***, A. Lahlou*, N. Talty, J. M. Manto, T. Zheng, K. L. Lam. *Seeing SDG 6 from Space: Local-Scale Monitoring of Piped Water and Sewage System Access Across Africa Using Satellite Imagery and Self-Supervised Learning*. **Water Research X** 100600 (2026). [Paper](#) | [arXiv](#).

*Equal contribution.

RESEARCH EXPERIENCE

Graduate Researcher

Mila – Quebec AI Institute, in collaboration with the Allen Institute for AI (AI2)

Jan. 2026 – Present

Montréal, QC

- Investigating why geospatial foundation models excel at semantic tasks such as crop-type classification yet lose their advantage on temporally driven targets—crop yield regression, wildfire forecasting, flood extent—a gap documented across recent EO benchmarks, where multi-temporal evaluation is thin and supervised baselines stay competitive.
- Fusing environmental drivers (climate reanalysis, fire-weather indices, agronomic covariates) into OlmoEarth (AI2; ~10 TB pretraining, up to ~300M parameters), whose pretraining spans satellite and sensor streams but not weather data. On CanadaFireSat (100 m, 8-day wildfire forecasting over Canada) and YieldSAT (12.2M yield samples at 10 m across four countries), adding environmental modalities to the pretrained backbone consistently beats satellite-only fine-tuning.
- Building toward a self-supervised pretraining objective that learns from environmental covariates alongside imagery—treating temporal and environmental structure as a pretraining signal rather than a downstream patch.

Machine Learning Research Intern

University Mohammed VI Polytechnic

Jul. – Dec. 2025

Benguerir, Morocco

- Assembled a 10-year crop-yield corpus with N/P/K fertilizer inputs across 150+ countries; built a transformer-based recommendation model and a retrieval-augmented agronomy assistant serving responses in ~2s.

Undergraduate Research Assistant

Duke Kunshan University

Jun. 2024 – Jun. 2025

Suzhou, China

- Curated a 300k-image ground-truthed satellite dataset spanning 50 of 54 African countries; applied DINO self-supervised pretraining to Sentinel-2/Landsat imagery to estimate piped-water and sewage access at 2.56 km resolution, reaching 96–97% classification accuracy and R^2 of 0.95/0.85 against UN Joint Monitoring Programme national statistics [Pub. 2] (*supervised by Dr. Ka Leung Lam*).

Last updated: September 2026

- Designed *SpeedTransformer*, a 0.73M-parameter encoder (RoPE, grouped-query attention, SwiGLU) that infers transport mode from speed alone—no coordinates or engineered features—achieving 95.97% on GeoLife versus 92.96% for the strongest prior transformer baseline, with cross-regional transfer from Swiss to Chinese data and validation in a 348-participant field deployment [Pub. 1] (*supervised by Dr. Charles Chang*).
- Prototyped *CarbonClever*, a WeChat mini-program for personal carbon-footprint tracking, used to collect the field-experiment dataset above.

Undergraduate Research Assistant

Jan. 2025 – Jun. 2025

University of Missouri

Columbia, MO

- Modeled wildfire spread with a Perceiver IO architecture, fusing satellite and climate modalities at differing spatial and temporal resolutions in a shared latent space without per-modality preprocessing (*supervised by Dr. Jordan Malof*).
- Built a VAE simulator over the learned latent space to generate counterfactual fire-progression scenarios, evaluated against the WSTS+ benchmark.

Undergraduate Research Fellow

Jun. 2023 – Aug. 2024

Duke University

Durham, NC

- Mapped invasive salt-marsh species (*Spartina alterniflora*) from Sentinel-2/Landsat time series using vision transformers, delivering landscape-scale outputs for coastal conservation planning (*supervised by Dr. Wenhong Li & Dr. Ding Ma*).
- Built a geospatial database on saltwater intrusion and sea-level rise across rural coastal plains, mining 1,000+ papers with BERT-based topic modeling and delivering an interactive ArcGIS dashboard (*supervised by Dr. Emily Bernhardt*).

TECHNICAL SKILLS

- **Programming:** Python (proficient, primary research language), SQL/Postgres (proficient), Java, C/C++, JavaScript, R (working)
- **ML/Data:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Matplotlib
- **Geospatial:** Google Earth Engine, ArcGIS, Sentinel-2/Landsat processing, rasterio, xarray
- **Systems/Tools:** Git, Docker, Linux, Google Cloud Platform, multi-GPU training (A100)
- **Web/Backend:** React, Node.js, Flask, FastAPI

GRANTS & AWARDS

- Dan Stefan Petrescu Fellowship, McGill University — \$30,000/year for two years (2026–2028)
- McGill Computer Science Graduate Award — \$30,000/year (2025); Graduate Registration Bonus — \$5,000 (2025)
- DKU Student Experiential Learning Fellowship — \$1,000 (2024); DKU Signature Work Research Grant — \$1,500 (2023); DKU Innovation Incubator — \$3,000 (2023)

SELECTED PROJECTS & PRESENTATIONS

- *Football AI Tracker*: end-to-end multi-object player tracking from low-quality broadcast recordings (with N. Talty).
- *Monitoring *Spartina alterniflora* Using Self-Supervised Learning*. Poster, **Duke Climate+ Symposium**, Durham, NC (2024).
- *Assessing Climate Change Risk of Rural Coastal Plains*. Poster, **Duke Climate+ Symposium**, Durham, NC (2023).

PROFESSIONAL EXPERIENCE

- **Data Analyst Intern**, Atos Morocco, Rabat (Oct.–Nov. 2022) — built real-time HR dashboards and scaled deployment across African branches to standardize reporting.
- **Data Analyst Intern**, XPerlean, Saint-Quentin, France (Jul.–Aug. 2022) — applied Faster R-CNN and YOLO to ceramic defect detection, raising accuracy from 70% to 85% and throughput by 10%.
- **Data Analyst Intern**, Al Jazeera Media Institute, Doha (Oct.–Dec. 2021) — processed 200k+ social media records via APIs for engagement analysis.

TEACHING, SERVICE & LEADERSHIP

- **Teaching:** Teaching Assistant, COMP 303 Software Design, McGill University (2025–Present); Mathematics & Computer Science Teaching Assistant, Duke Kunshan University (multiple terms, 2022–2025); Partner & Mathematics Tutor, FADI Academy.
- **Service:** Lab Representative, Mila – Quebec AI Institute (2025–Present); Board Member, Muslims in Tech; Software Developer, DKU Computer Science Club ([HackDKU 2025](#) website).
- **Leadership:** Resident Assistant and Varsity Soccer Co-Captain, Duke Kunshan University; Team Leader, Spectator Services Volunteer, FIFA World Cup 2022, Qatar.

LANGUAGES

Arabic (native); French (fluent); English (fluent); Chinese (intermediate); Spanish (conversational).